Seva First Innovation Challenge (SFIC) | Concept Document

AgriGuard

Early Crop Disease Screening and Safe Advisory for Small Farmers

A phone-based tool that helps farmers spot crop disease, pest damage and plant stress early, understand how serious it is, and know safe next steps — or when to call an expert.

Theme	**Samriddh Annadata, Samriddh Bharat** (Prosperous Farmers, Prosperous Bharat)
Sub-theme	Agricultural Innovation
Category	Undergraduate and Above
Track	Track 1: Product / Technology Solution
Development stage	Prototype
Date	October 2026

1 Problem Statement

Small and medium farmers in India cannot identify crop diseases and pest attacks early or get timely, reliable advice, so outbreaks spread unchecked and destroy a large share of their harvest and income.

30–35%
of India's annual crop yield lost to pests (ICAR)

<1 : 5,000
extension workers to farmers; norm 1:750–1:1,100 (NAAS/ICRISAT)

₹1,125 cr
cotton relief, Rajasthan 2023 — paid after the crop was lost

₹322 cr
soybean yellow-mosaic relief to 4.95 lakh MP farmers, 2025

The problem

Crop diseases and pests are among the biggest causes of income loss for Indian farmers. The core issue is **late or wrong identification**: by the time a farmer is sure what is wrong, the damage has spread. Most small farmers identify problems by eye, from neighbours or from the local pesticide shop, because expert help is scarce. Farmers then often spray several chemicals that do not work, adding cost and harming soil, health and the environment.

How often it happens — five recent Indian outbreaks

Crop · Pest/disease	Where · When	What happened	Source
Cotton · Pink bollworm	Punjab, Haryana, Rajasthan · 2023	~65% loss in Punjab & Haryana, up to 90% in parts of Rajasthan; farmers with debts of ₹6–8 lakh; losses recurring since 2017.	Down To Earth, 19 Oct 2023
Chilli · Black thrips	Telangana, Andhra Pradesh · 2021–22	At least 3.6 lakh ha hit; yield fell from 5–6 t/ha to ≤2.5 t/ha. "I used at least six different pesticides but nothing worked."	Down To Earth, 12 May 2022
Coconut · Root wilt & whitefly	Coimbatore & Tirupur, Tamil Nadu · 2024–25	Yield down 30–40% in the Pollachi belt; 1.9 lakh acres affected in TN; trees cut down; pests return ~10 days after spraying.	DT Next, 2024; The Quint, 2025
Banana · Panama wilt (TR4)	Bihar, Uttar Pradesh · 2020	Over 11,000 ha infected; farmers' earnings fell to less than half; ₹12 lakh spent on a single farm.	Down To Earth, 21 Sep 2020
Soybean · Yellow mosaic virus	Madhya Pradesh · 2025	Severe outbreak in Mandsaur; damage survey ordered after infection; ₹322 crore relief to 4,94,605 farmers.	Global Agriculture; LatestLY, 2025

Common pattern: late or wrong identification → no timely expert guidance → money wasted on ineffective sprays → relief that arrives only after the harvest is lost. It repeats every season, in every major crop and region.

2 Who Is Affected

- **Small and marginal farmers**, often on leased land and borrowed money, for whom one outbreak means debt.
- **Growers of disease-prone crops** — cotton, chilli, coconut, banana, soybean, tomato, potato and more.
- **Farm labourers**, whose work and wages fall when crops fail.
- **Women farmers and rural households** with even less access to extension services.
- **Extension officers and KVK staff**, who cannot reach thousands of farmers each season.
- **State governments and consumers**, through large relief payouts and higher prices.

3 Proposed Solution

AgriGuard is a mobile-first web application. The farmer photographs an affected leaf with an ordinary smartphone — choosing the crop or letting AgriGuard detect it — and within seconds receives a screening result, how confident the system is, how serious the visible damage is, and safe next steps in their own language.

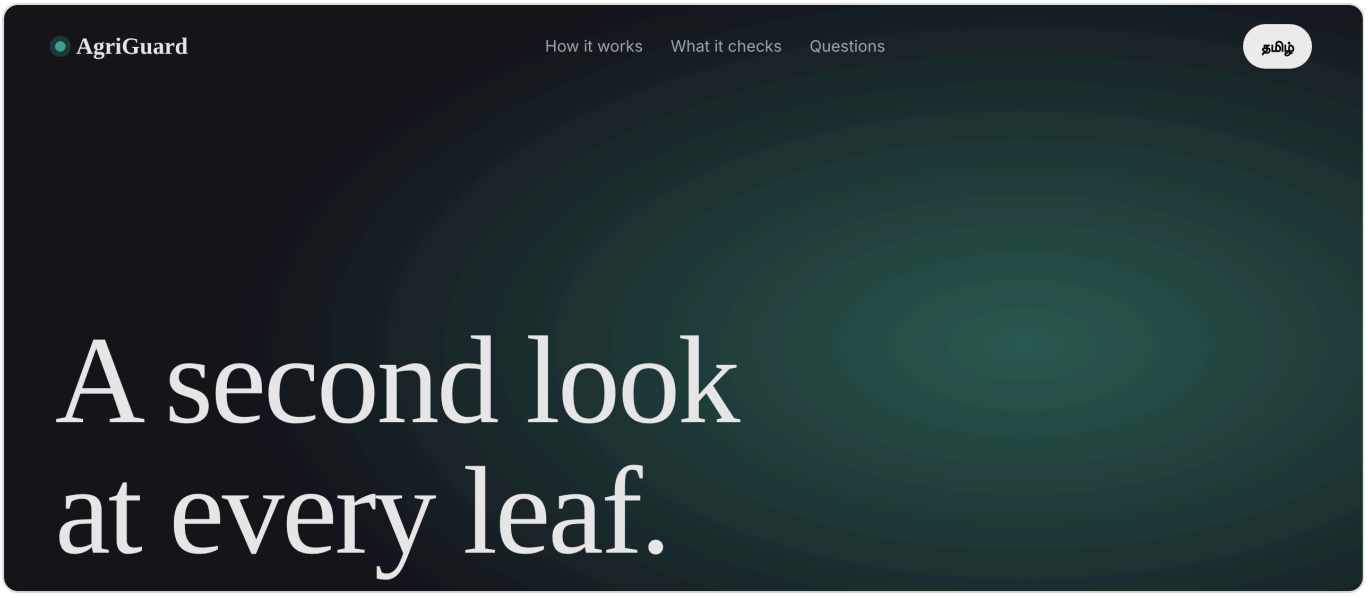
What the farmer gets

- Likely problem or "Healthy", with an honest confidence level.
- **"Not sure"** when the photo is unclear or confidence is low — with advice to retake it or contact the local agriculture officer/KVK.
- Severity estimate: **Low / Moderate / High** from the visible affected leaf area.
- Advisory: immediate actions, symptoms to confirm, prevention, low-risk organic and cultural practices, and when to seek help.
- Scan history and before/after comparison; dashboard of trends.
- English and Tamil with an instant switch; private accounts.

What it covers

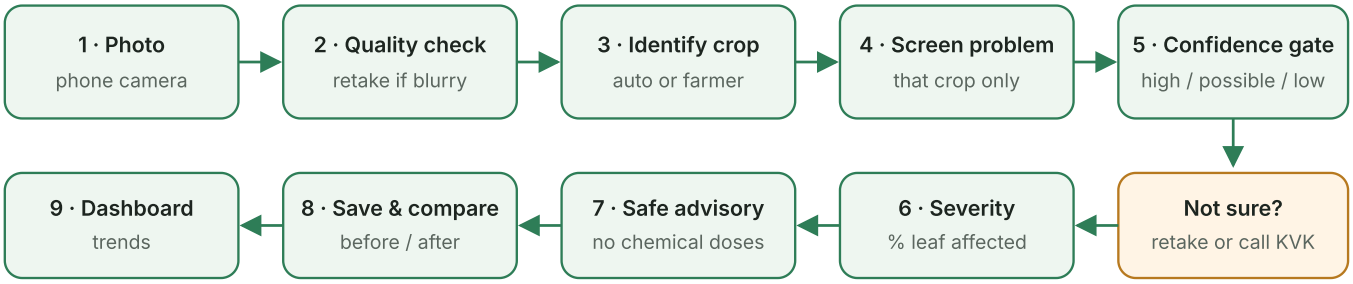
- Fungal, bacterial and viral diseases
- Pest damage (e.g. thrips, fall armyworm, borers)
- Nutrient deficiencies (e.g. nitrogen, potassium)
- Water and heat stress
- Designed for 14 common crops: rice, wheat, maize, cotton, banana, mango, grape, tomato, potato, chilli/pepper, brinjal, onion, cabbage, cucumber

Safe by design: no pesticide brand names or doses — enforced by an automated test. Every advisory ends with a referral to the local agriculture officer or Krishi Vigyan Kendra (KVK).



AgriGuard interface design — landing page with Tamil language switch.

4 How the Solution Works



Optional field sensing
ESP32 kit with soil-moisture, temperature and humidity sensors sends readings every minute.
Shown as context (e.g. humid, wet weather favours fungal disease), never as a diagnosis.

Step-by-step flow from photo to advisory.

Technology

Layer	What is used
Farmer app	Next.js progressive web app — runs in a phone browser, no app-store install; client-side image compression for slow rural networks; Android wrapper app
Backend	Python FastAPI; photo-quality checks; confidence rules; severity by image processing (leaf vs lesion pixels)
AI model	Deep-learning image classifier (transfer learning on public, licence-checked crop-disease datasets) behind a replaceable adapter — a better or field-trained model can be swapped in by configuration
Data	SQL database for accounts, scans and sensor readings; separate private image storage per user; advisory stored as editable data (English/Tamil)
Field hardware (optional)	ESP32 + capacitive soil-moisture sensor + temperature/humidity sensor; simulator for demonstrations

5 What Makes It Innovative

AgriGuard	Typical approach today
Says "Not sure" and refers to an expert when confidence is low	Photo apps always return a disease name, even when wrong
Crop-aware screening — crop identified first, then only its problems compared	One model guesses across all crops, causing confusion
Severity + tracking — % of leaf affected, before/after comparison	A label only, no sense of how serious or whether it is spreading
Beyond disease — pests, nutrient deficiency, water/heat stress	Disease-only tools; deficiencies are often mistaken for disease
No chemical prescriptions, enforced by test; KVK referral on every advisory	Shop-led advice; farmers spray several chemicals by guesswork
Expert-editable knowledge — advisory stored as data	Advice hard-coded or copied from the internet
Built for rural India — browser-based, Tamil first, low bandwidth	English-only apps needing installs and fast internet

6 Development Stage and Evidence

Stage: Prototype. The full application is built and runs end to end. The image model is currently a **placeholder used to test the flow**; connecting and evaluating a trained model on real field photos is the next step.

Completed

- Problem study from recent Indian outbreak reports and national data
- Accounts, crop selection / auto-detect, photo upload with quality checks
- Result page with confidence and severity; "Not sure" handling
- English & Tamil advisory for the first 10 disease classes (tomato, potato, pepper, cucumber)
- History, before/after comparison, dashboard
- ESP32 firmware and sensor simulator; Android wrapper app

Remaining

- Connect and evaluate a trained model
- Real field-photo test set from local farms; report accuracy honestly (lab vs field)
- Expert review of advisory; extend to more crops
- Testing on low-end phones and slow networks
- Pilot with 50–100 farmers through a KVK or FPO

Evidence: automated testing

All **30 automated backend tests pass** (run on the current code, October 2026), including:

Area	What the tests check
Photo quality	Valid image accepted; too dark, too bright, invalid or oversized files rejected
Confidence rules	High, possible and uncertain cases; small-margin cases withheld; display capped at 99%
Severity	Healthy gives none; every severity is flagged as an estimate
Advisory safety	No chemical brand names; KVK referral present; Tamil falls back to English when missing
Full flow	Login and scan, comparison, health check, sensor data validation and device authentication

7 Expected Impact

- **Economic:** earlier action protects yield and income; fewer wasted sprays cut input costs.
- **Social:** same-day guidance instead of waiting for an officer; less stress and debt after crop failure; inclusive of low-literacy and women farmers.
- **Environmental:** no-prescription advice reduces unnecessary chemicals, protecting soil, water, pollinators and farmworkers.
- **Institutional:** KVKs reach more farmers; consented, anonymised scan data can flag outbreaks early so relief and advice are targeted before losses.

8 Estimated Cost

Built with open-source tools and free cloud tiers. Figures are approximate and will be confirmed during the pilot. **Free for farmers** on their own smartphone.

Development (one-time)	Approx. (₹)
Software, frameworks, datasets (open-source)	0
Model training on cloud GPU	0 – 5,000
Field photo collection & travel	5,000 – 10,000
ESP32 sensor kit (optional, per kit)	1,500 – 2,500
Domain & miscellaneous	1,000 – 2,000
Total	**≈ 10,000 – 20,000**

Operation (per year, pilot scale)	Approx. (₹)
Hosting: app, database, images	0 – 15,000
Advisory review & translation	10,000 – 20,000
Maintenance & model updates	Student team
Total	**≈ 10,000 – 35,000**

At scale, costs can be met by agriculture departments, KVKs, FPOs or CSR partners — not farmers.

9 Implementation and Scalability

Phase	What happens
1 · Pilot (3–6 months)	Trained model for 3–5 common local crops; real field-photo testing; pilot with 50–100 farmers through a KVK or FPO with expert review of results
2 · District	More crops based on pilot demand; more regional languages; extension officers and FPO volunteers onboard farmers; consented photos and expert feedback improve accuracy

3 · State / national	Partnerships with state agriculture departments and agricultural universities for region-specific crops and advice; cloud hosting scales with users; no app-store install needed
Future features	Offline mode for low-connectivity villages; voice output; SMS/WhatsApp summaries; anonymised disease maps that alert officers when many scans in one area show the same problem

10 Potential Adopters

- Krishi Vigyan Kendras (KVKs) under ICAR
- State agriculture & horticulture departments (e.g. Tamil Nadu Department of Agriculture)
- Agricultural universities and ICAR institutes (e.g. TNAU) — validation and expert-labelled images
- Farmer Producer Organisations (FPOs) and cooperatives
- ATMA and Common Service Centres for village-level reach
- NABARD and rural development programmes
- Agri-focused NGOs and CSR foundations
- Government digital agriculture platforms (as a screening module)

11 Support Required

- **Mentorship** from plant pathologists and agriculture scientists (KVK, agricultural university or ICAR) to review disease classes and advisory content.
- **Data access:** expert-labelled field images of Indian crops and permission to photograph farms through a KVK or FPO.
- **Technical mentorship** on running models on low-end phones and slow networks, including offline use.
- **Cloud credits** for training, hosting and image storage during the pilot; **seed funding** of about ₹20,000–50,000 for field data, sensor kits, travel and translation review.
- **A pilot partner** (KVK, FPO or agriculture department) to test with 50–100 farmers; help with Tamil and other regional-language content; guidance on data privacy, IP and incubation.

12 Responsible Use

AgriGuard is a **screening and decision-support tool, not a guaranteed diagnosis**. Results are labelled as estimates, low-confidence cases are referred to experts, no chemical brands or doses are given, sensor readings are shown only as context, and farmers' photos are used for their own history unless they agree otherwise.

13 References

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